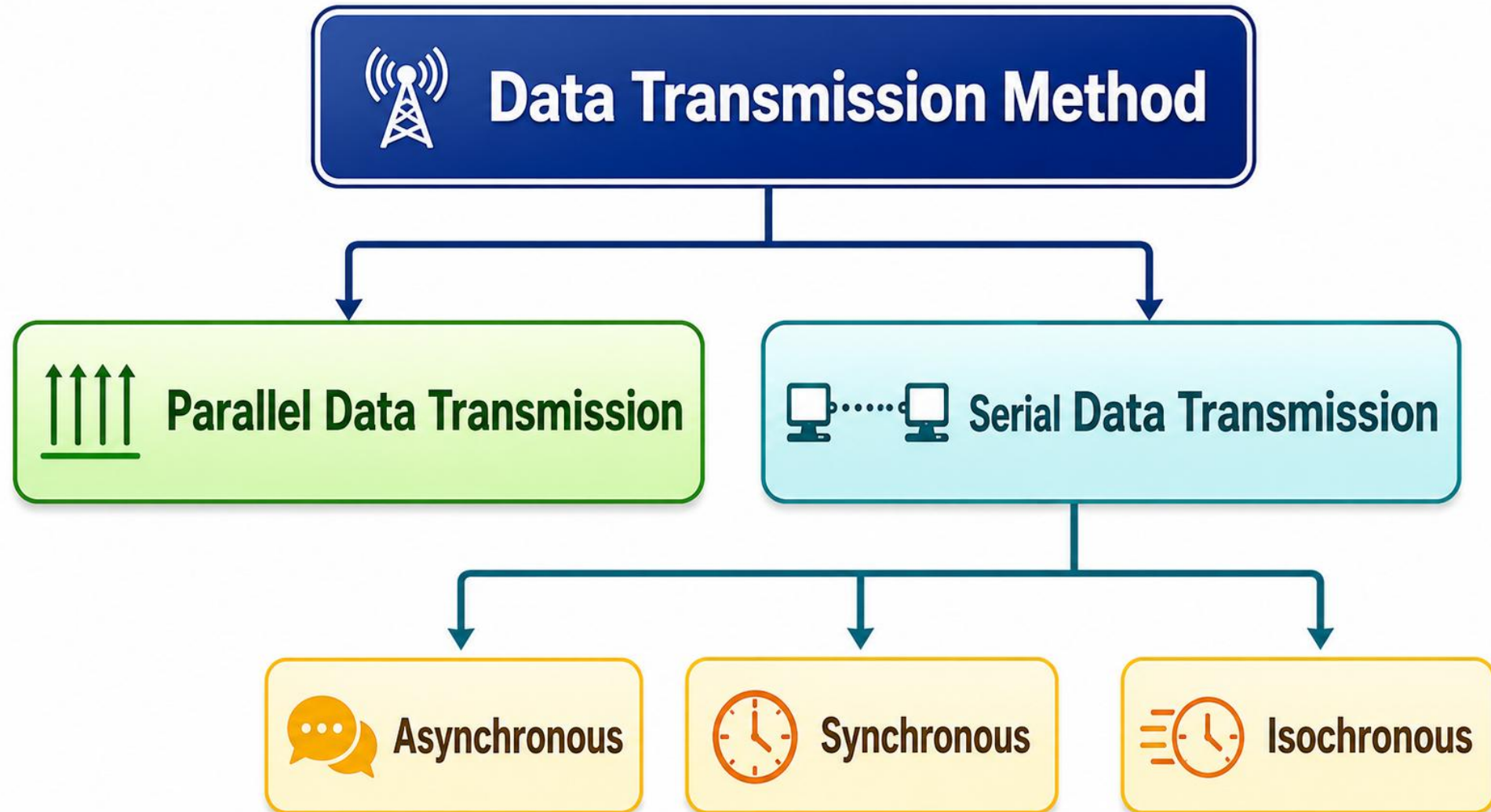




Types of Data Transmission Methods





Parallel Data Transmission



Multiple bits are sent simultaneously through multiple channels/wires.



8 bits move together

- ☐ Used inside computer circuits.
- ☐ Used between nearby devices.
- ☐ Example: old printer parallel port.



Parallel Transmission: Advantages and Disadvantages



Advantages

- ✓ Fast
- ✓ Many bits move simultaneously
- ✓ Good efficiency for short distance
- ✓ Data moves word by word

Disadvantages

- ☐ Not suitable for long distance
- ☐ Needs many wires
- ☐ More costly
- ☐ Wire/clocking problems increase with distance



Serial Data Transmission



Bits are sent one after another through a single channel/wire.



- ☐ Used for remote/long-distance communication.
- ☐ Needs fewer wires than parallel transmission.
- ☐ Examples: USB, modem, mouse, serial communication devices.



Parallel vs Serial Data Transmission



Point	Parallel	Serial
Bits sent	Many bits at the same time	One bit at a time
Channel/wire	Multiple channels/wires	Single channel/wire
Distance	Short distance	Longer distance
Cost	Costly for long cables	Cheaper for long distance
Example	Computer bus, old printer port	USB, modem, mouse

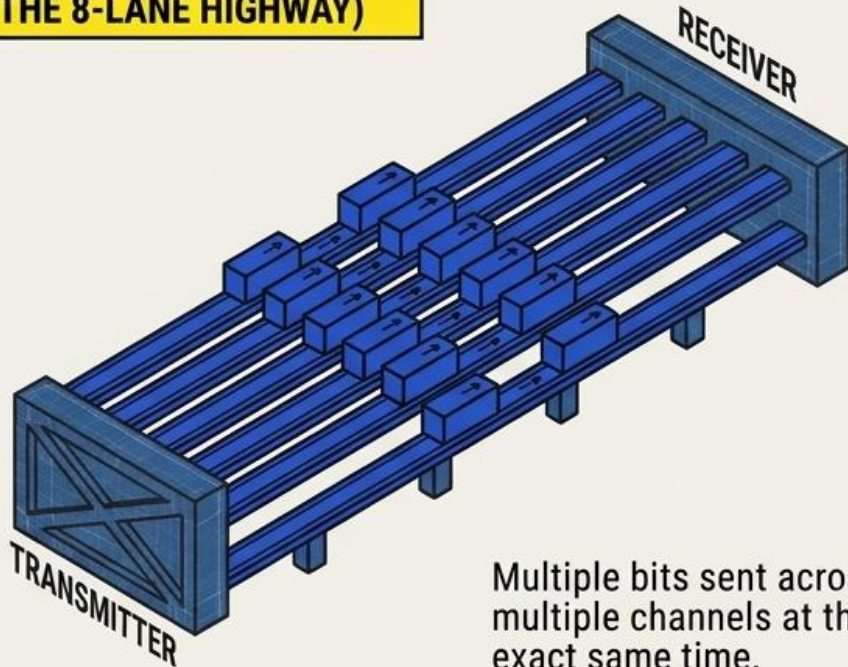


Parallel vs Serial Data Transmission



THE HIGHWAY VS. THE FAST LANE

PARALLEL TRANSMISSION (THE 8-LANE HIGHWAY)

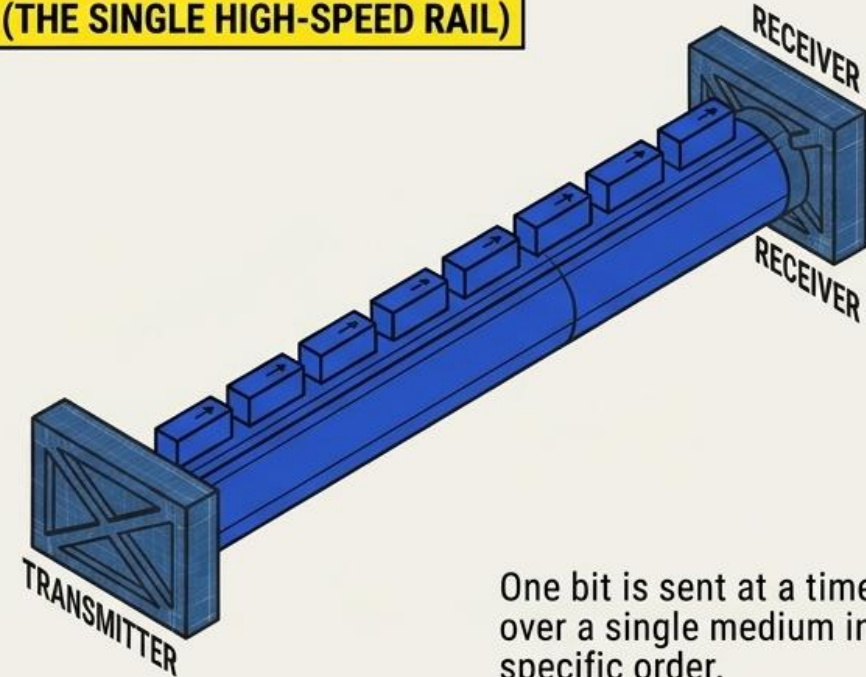


Multiple bits sent across multiple channels at the exact same time.



Commonly used in traditional printers and internal computer circuits.

SERIAL TRANSMISSION (THE SINGLE HIGH-SPEED RAIL)



One bit is sent at a time over a single medium in a specific order.



Commonly used in USBs, computer mice, and modems.



Quick Check: Parallel or Serial?



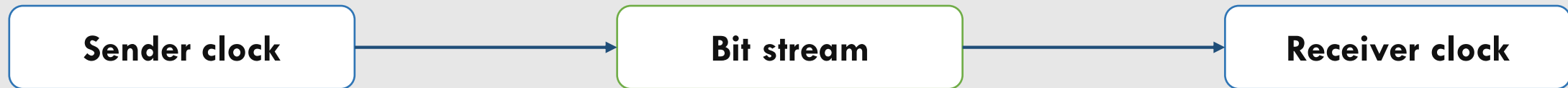
Scenario	Answer
Data sent inside CPU bus	Parallel
USB keyboard sends key data	Serial
Old printer parallel port	Parallel
Modem sends data to network	Serial
Many bits over many wires	Parallel



Bit Synchronization



Bit synchronization means matching timing between sender and receiver so each bit can be detected correctly.



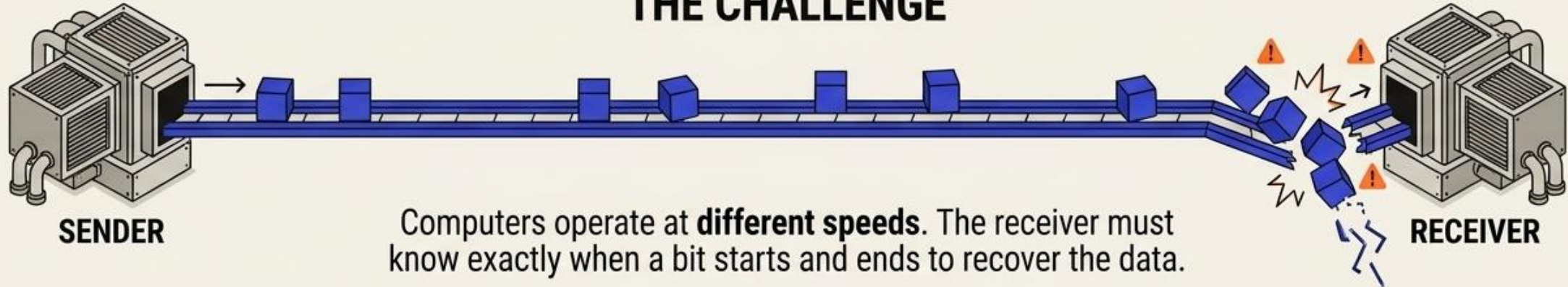
- ☐ The receiver must know where one bit starts and where it ends.
- ☐ If timing is wrong, the receiver may read wrong bits.
- ☐ Synchronization is especially important in serial transmission.



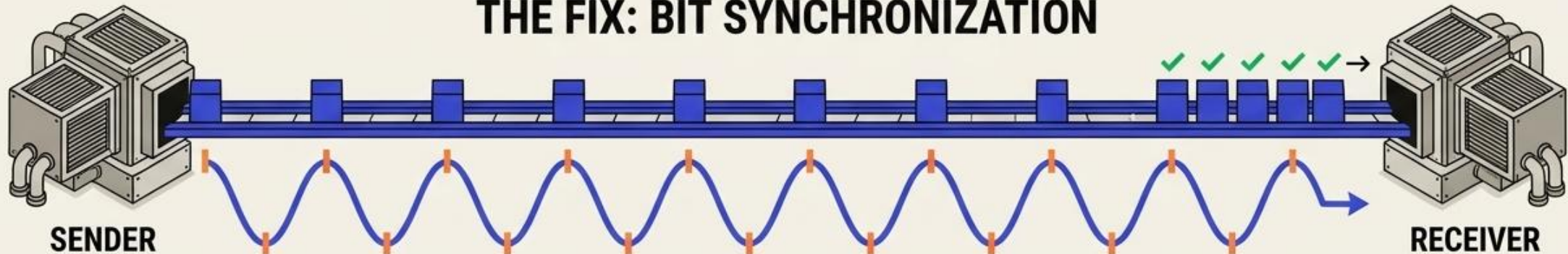
Bit Synchronization



THE CHALLENGE

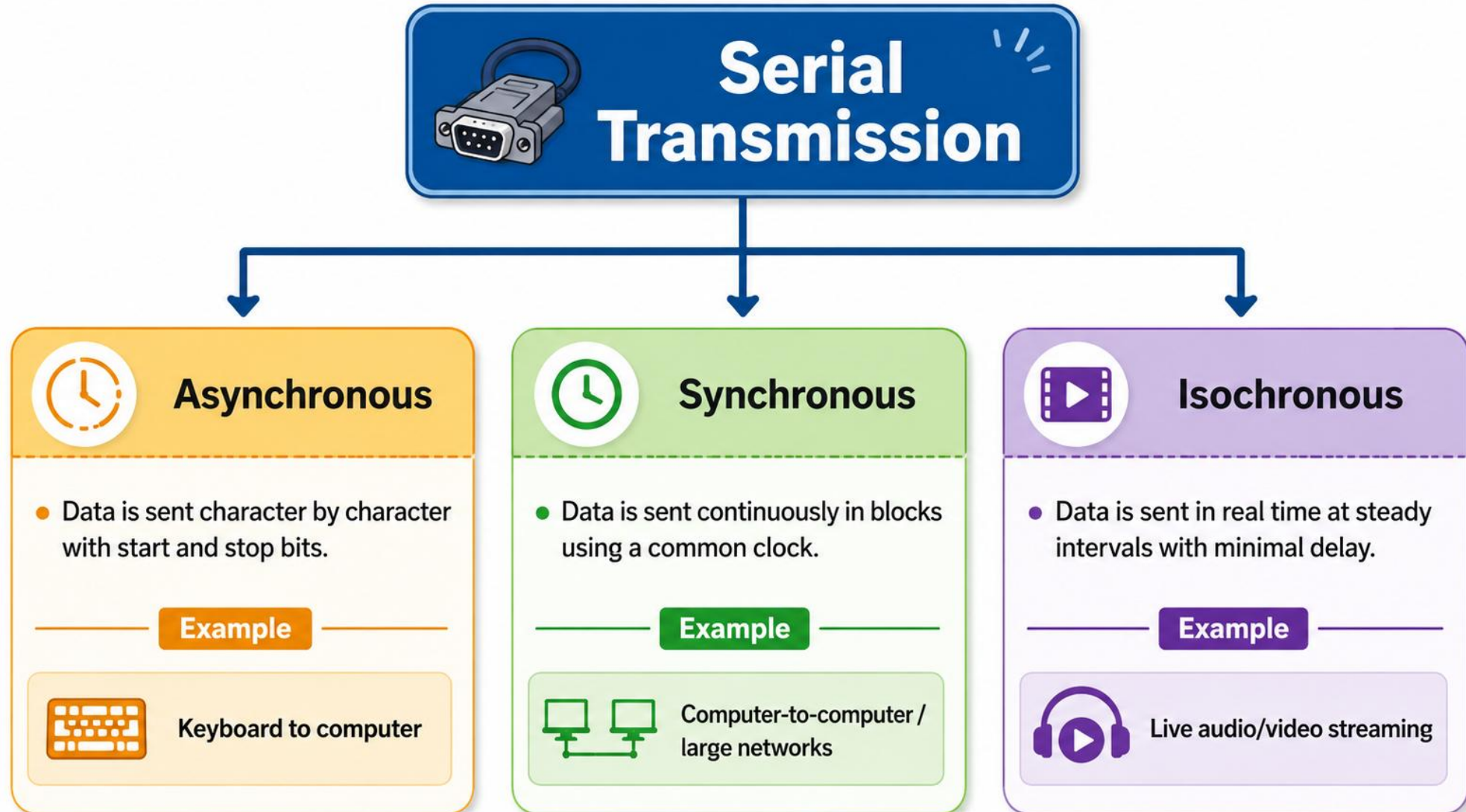


THE FIX: BIT SYNCHRONIZATION





Types of Serial Transmission





Asynchronous Transmission



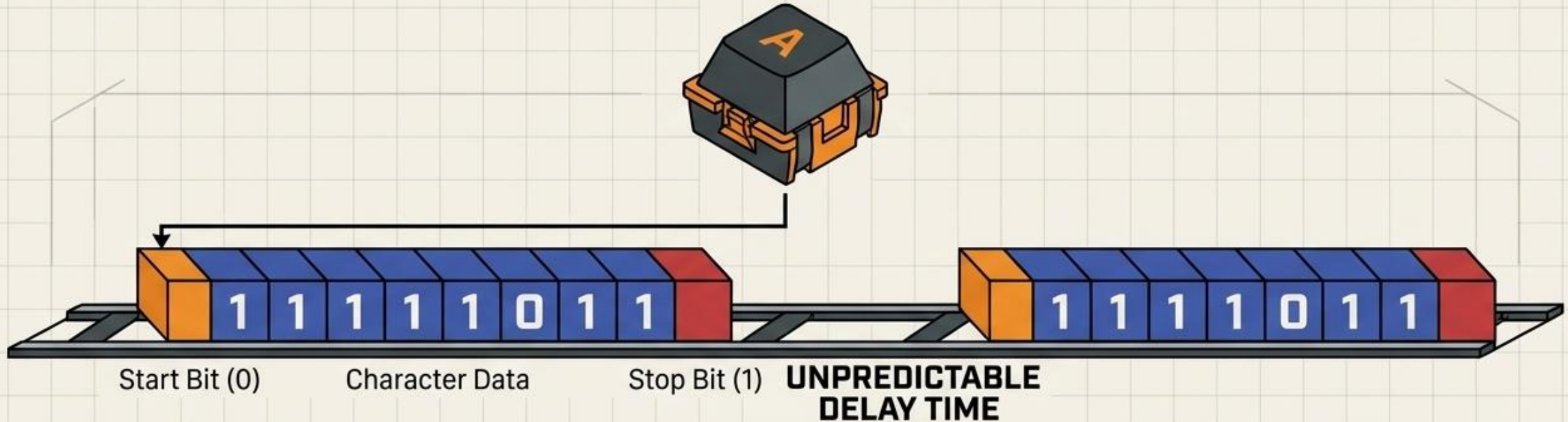
Data is transmitted character by character. Each character uses start and stop bits.



- ☐ Sender can send data at any time.
- ☐ Receiver is always prepared to receive.
- ☐ Suitable for small amounts of data at slower speed.



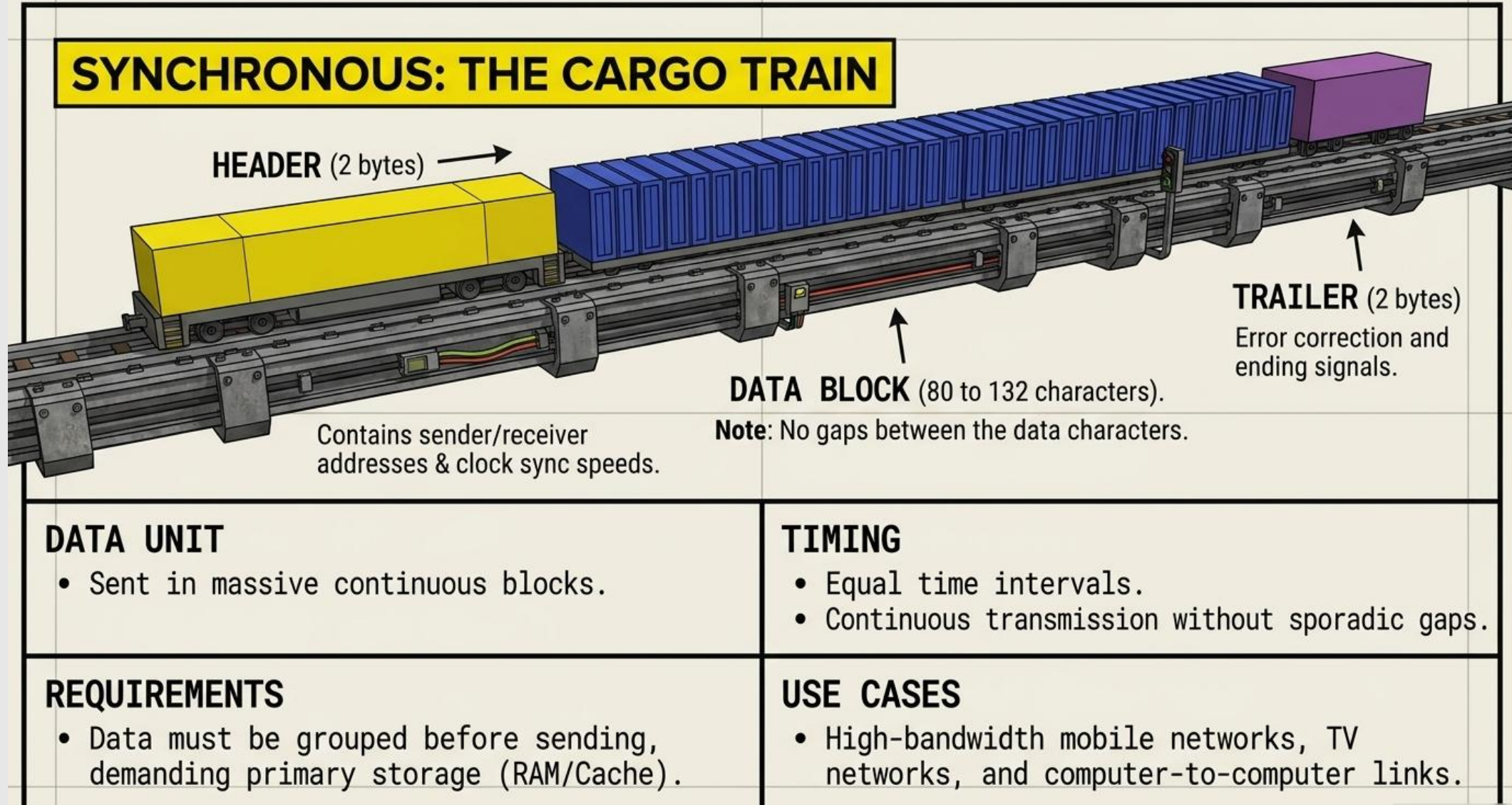
Asynchronous Transmission



DATA UNIT	TIMING & GAPS	STORAGE
• Sent character by character (8-bit). • Total package becomes 10/11 bits.	• Unequal time intervals. • Sender can transmit at any time. • Unpredictable gaps exist between characters.	• No primary storage (RAM) needed at the sender's end.



Synchronous Transmission

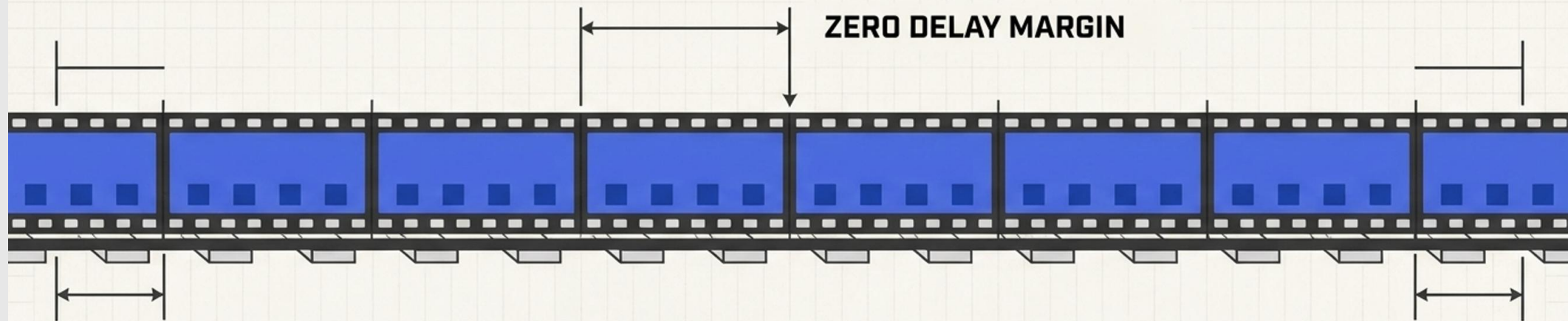




Isochronous Transmission



ISOCHRONOUS: THE LIVE FEED



The Hybrid Approach: A cross between Synchronous (block format) and Asynchronous (uses start/stop bits).



The Golden Rule: The time interval between data blocks must remain near zero.



No Looking Back: There is no time to retransmit lost data. If a packet is lost, it stays lost.



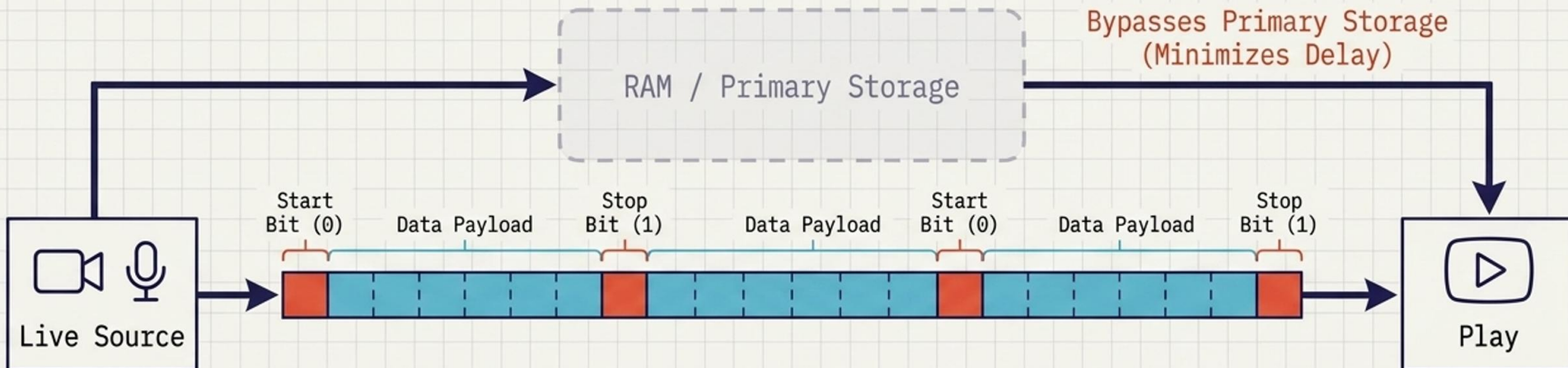
Real-World Application: Essential for real-time streaming, live TV broadcasts, and video conferencing where lag ruins the experience.



Isochronous Transmission



Isochronous: The Real-Time Hybrid



The Mechanism	The Bumpers	Architecture Note
A cross between Synchronous and Asynchronous methods. Transmits data in blocks with almost zero interruption, utilizing start/stop bits instead of pure synchronization.	Bypasses primary storage to keep time delays strictly at a minimum (nearly zero). Because it cannot wait, errors cannot be retransmitted or checked.	Unreliable for perfect data integrity, but absolutely essential for real-time applications where speed outweighs a dropped packet: live TV broadcasts, streaming video, and voice calls.



Asynchronous vs Synchronous vs Isochronous



Point	Asynchronous	Synchronous	Isochronous
Unit	Character	Block / frame	Real-time blocks
Timing	Unequal gaps allowed	Equal timing	Minimum delay
Storage	Not required	Required	Usually not required
Speed	Slower	Fast	Real-time
Example	Keyboard	Large networks	Video call / streaming

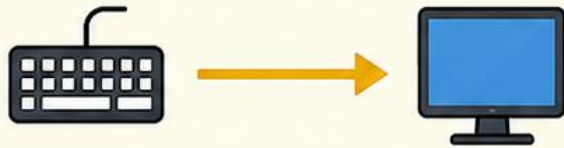


Data Transmission Modes



Direction of data flow between two devices

1. SIMPLEX



DEFINITION

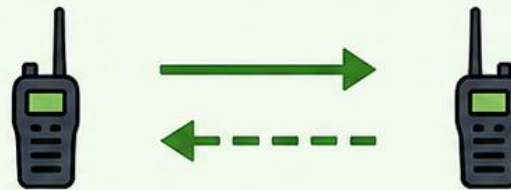
One-way communication only



EXAMPLES

Keyboard to computer, radio, TV

2. HALF-DUPLEX



DEFINITION

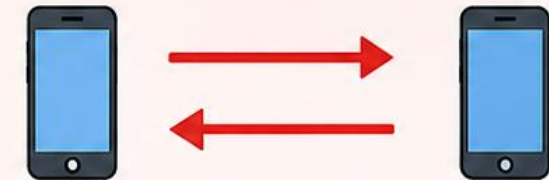
Two-way communication, but not at the same time



EXAMPLES

Walkie-talkie, fax, SMS

3. FULL-DUPLEX



DEFINITION

Two-way communication at the same time



EXAMPLES

Telephone, mobile phone, video call



● **Simplex:** one-way only

● **Half-duplex:** two-way, but not at the same time

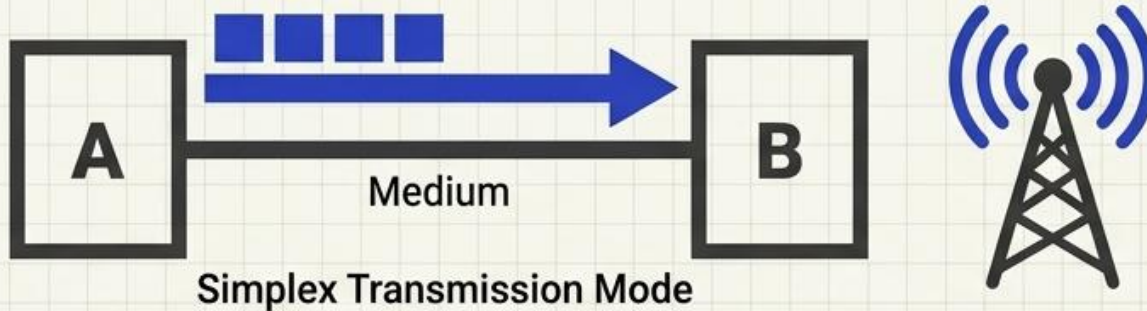
● **Full-duplex:** two-way at the same time



Simplex and Half Duplex



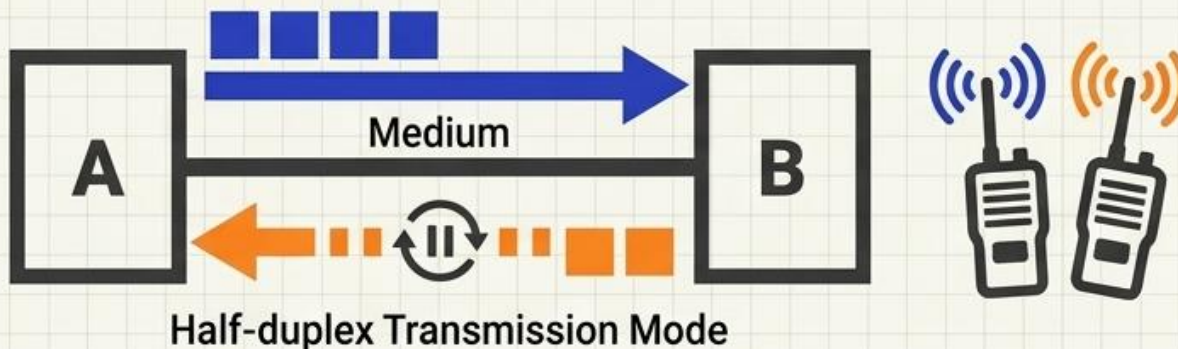
SIMPLEX MODE (THE MEGAPHONE)



Transmitter only sends; receiver only receives. Uses full bandwidth in one direction.

Examples: Keyboards, Radio broadcasting, TV networks.

HALF-DUPLEX MODE (THE WALKIE-TALKIE)



Both devices can send and receive, but not at the same time. When one transmits, the other waits.

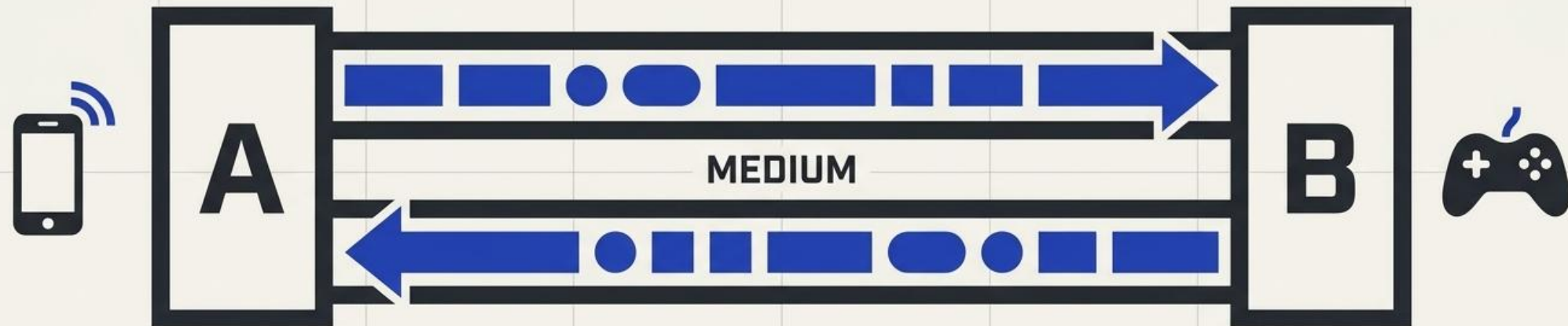
Examples: Walkie-talkies ("Over!"), Fax machines, SMS.



Full Duplex



THE TWO-WAY HIGHWAY: FULL-DUPLEX



SIMULTANEOUS ACTION:

Both ends can send and receive data at the exact same time.

NO WAIT TIMES:

The channel is split into two dedicated paths, eliminating data collision entirely.

PERFORMANCE:

The highest performing and fastest communication mode available.

EXAMPLES:

Telephones, mobile networks, and multiplayer online gaming.



Simplex vs Half-Duplex vs Full-Duplex



Point	Simplex	Half-Duplex	Full-Duplex
Direction	One direction	Both directions, not same time	Both directions same time
Communication	One-way	Two-way in turns	Simultaneous two-way
Sender/Receiver role	One sender only	Both can work, not same time	Both can work at same time
Performance	Lowest	Better than simplex	Highest
Example	Radio, TV, keyboard	Walkie-talkie, SMS	Phone call, mobile network



Activity: Identify the Mode



Example	Mode
Teacher speaks through microphone to students	Simplex
Walkie-talkie conversation	Half-duplex
Mobile phone call	Full-duplex
Keyboard sends input to computer	Simplex
SMS exchange between two people	Half-duplex

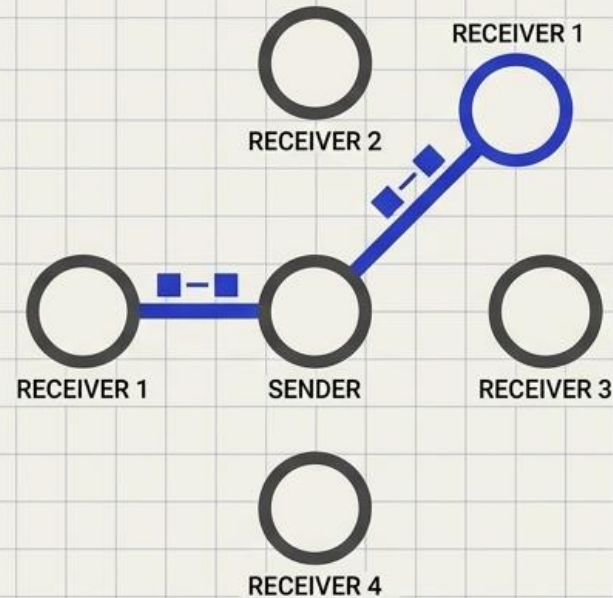


Data Distribution Modes



DELIVERY ROUTES: DATA DISTRIBUTION MODES

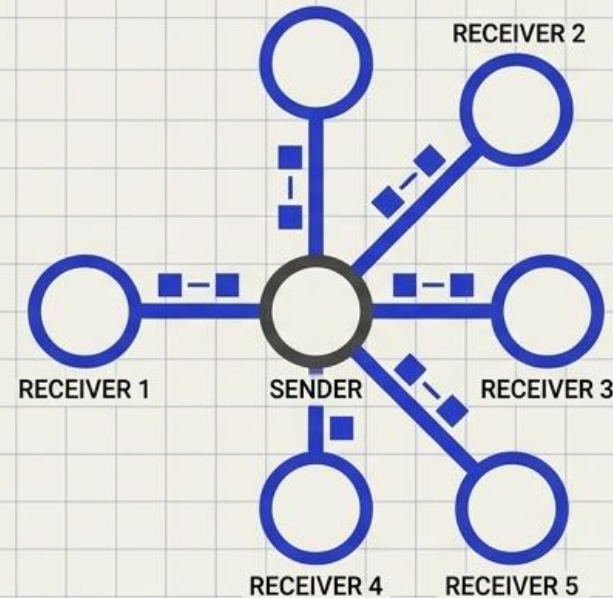
UNICAST (1-to-1)



Private communication.
Examples: Pager, private SMS,
Walkie-talkie.



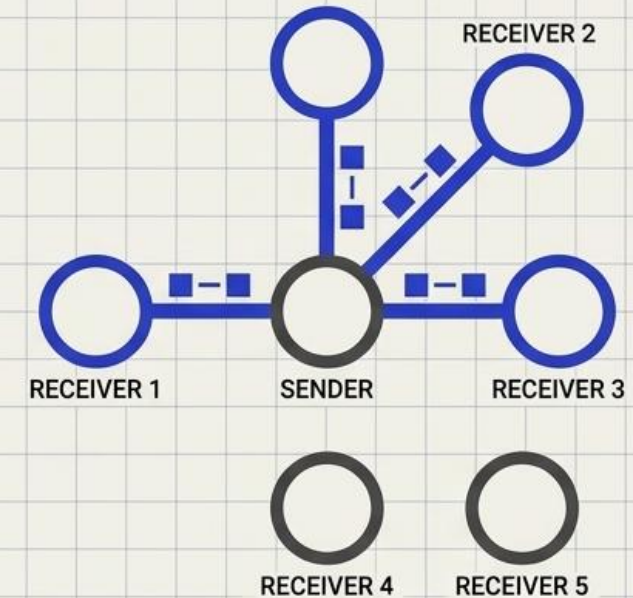
BROADCAST (1-to-All)



Total network saturation.
Examples: Radio, traditional
television expansion.



MULTICAST (1-to-Many)



Permission-based group delivery.
Examples: Group WhatsApp chats,
Video conferencing.





Data Distribution Modes



Point	Unicast	Broadcast	Multicast
Receiver	One specific receiver	All receivers	Selected group
Pattern	One to one	One to all	One to many selected
Access	Only target receives	Everyone receives	Only permitted users receive
Example	Single SMS	Radio / TV	Video conference

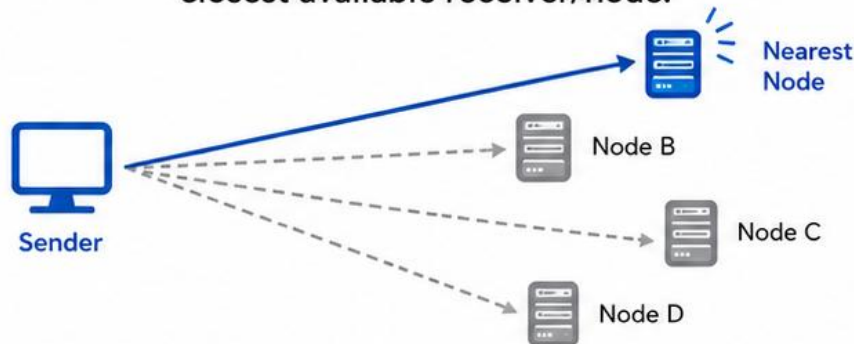


Advanced Data Distribution Modes



Anycast

Data is sent to the nearest or closest available receiver/node.



- Closest node receives the data
- Fast and efficient
- Useful for large/global networks



Examples

DNS servers,
CDN, cloud services



Geocast

Data is sent to all receivers inside a specific geographic area.



- Works within a defined area
- Location-based delivery
- Useful for regional services



Examples

Local alerts,
area-based messages,
regional ads

Target	Anycast: one nearest node	Geocast: all nodes in an area
Coverage	Anycast: nearest path	Geocast: geographic region
Best use	Anycast: speed	Geocast: area-based delivery



Remember: **Anycast** = nearest one, **Geocast** = everyone in a selected area.



Data Distribution Modes



SYNTHESIS: THE ANATOMY OF A GROUP VIDEO CALL

SERIAL TRANSMISSION

Data travels bit-by-bit over the wireless network to span long distances efficiently.



ISOCHRONOUS

Video frames are transmitted continuously with a near-zero delay margin to prevent lag.



FULL-DUPLEX

You can talk and hear your friends talking at the exact same time without the audio cutting out.



MULTICAST

Data is delivered only to the specific friends granted permission in the chat, not broadcasted publicly.



GROUP CALL

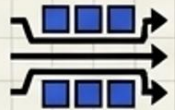




Summary



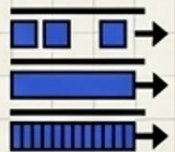
1. METHODS (GROUPING)



- **Parallel:** Fast, multi-lane, internal.



- **Serial:** Long-distance, single-lane.



- **Subsets:**
 - **Asynchronous** (Sporadic gaps)
 - **Synchronous** (Massive continuous blocks)
 - **Isochronous** (Live, zero-delay)

2. MODES (DIRECTION)



- **Simplex:** One-way only.



- **Half-Duplex:** Two-way, taking turns.

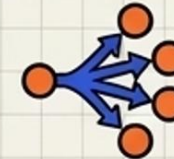


- **Full-Duplex:** Two-way, simultaneous action.

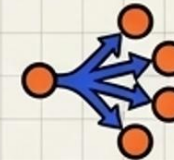
3. DISTRIBUTION (RECIPIENTS)



- **Unicast:** 1-to-1 (Private).



- **Broadcast:** 1-to-All (Public saturation).



- **Multicast:** 1-to-Many (Permission groups).